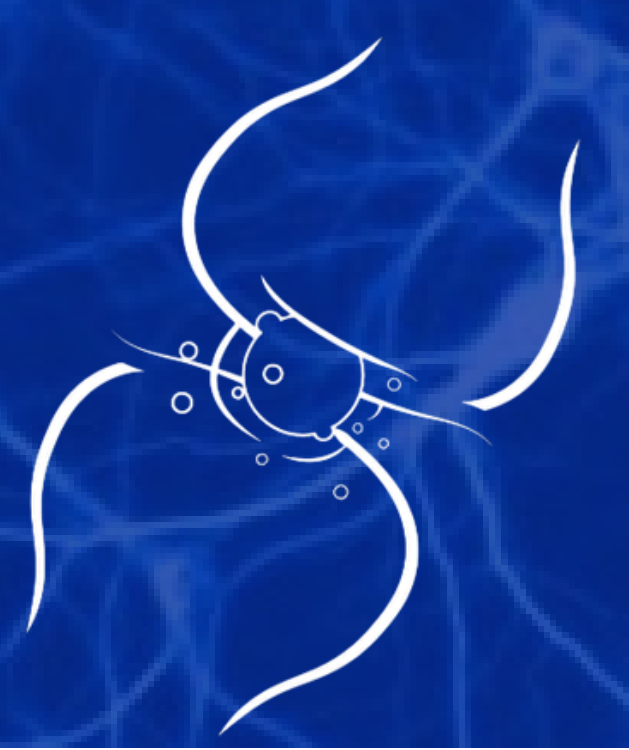




Non-invasive and Adaptive Brain-Computer Interface Based on Electroencephalographic Signals from the Primary Motor Cortex



Abstract

Brain-Computer Interfaces (BCIs) represent one of the most innovative frontiers in the field of applied Neurology. The engineering interest surrounding this type of device is mainly focused on the human-device connection and the analysis of electroencephalographic (EEG) signals, finding applications in the treatment of several neuropathologies that may impair an individual's voluntary movement. Among the main clinical conditions addressed by the applicability of the present project are Locked-in Syndrome (LS), Amyotrophic Lateral Sclerosis (ALS), multiple sclerosis (MS) and types of neurotrauma capable of causing irreversible damage to the spinal cord. The development of an integrated environment, combining hardware and software components, therefore enables the creation of new communication modalities, placing versatility, practicality and patient well-being at the core of system use.

Introduction

Pathologies and traumatic injuries affecting the CNS or spinal cord can disrupt communication between the brain and voluntary muscles, although brain activity often remains partially or fully intact. Brain-Computer Interfaces offer a promising alternative communication pathway, particularly through signals acquired from the Primary Motor Cortex (M1), involved in motor command generation. This project develops a non-invasive, adaptive EEG-based BCI for real-world use in scenarios where M1 functionality is preserved, integrating dedicated software for acquisition, analysis, and training, with attention to usability in clinical and rehabilitation settings.

The clinical challenge

Amyotrophic Lateral Sclerosis, Locked-in Syndrome, and spinal cord injury disrupt the pathway between motor intention and its physical execution, while movement-related cortical activity often remains partially or fully accessible. The system therefore targets the M1, which governs voluntary movement through the pyramidal system and from which such activity can be recorded via EEG. Event-Related Potentials, particularly the P300 wave, are studied at this level for their correlation with physical and imagined movement.

Hardware and Signal Acquisition

The hardware comprises two parts: the applied part and the processing board. Hypoallergenic polyester straps replace the conventional EEG cap, positioning dry Ag/AgCl electrodes (Cz, C1/C2, C3/C4, FC3/FC4) over the M1, with two Ag/AgCl reference electrodes at the ipsilateral and contralateral earlobes. Snap cables, twisted to reduce interference, connect to the OpenBCI Ganglion, a 4-channel PCB with an AD8237 amplifier, positioned at the C7 vertebra or ipsilateral biceps.

Software and Processing Architecture

The board transmits acquired EEG signals to the software, organised into two procedures. The first covers signal study and user training. The OpenBCI GUI enables real-time visualisation for signal quality, while MATLAB, via EEGLAB and ERPLAB, analyses the data and identifies ERP features. Training is developed in MATLAB App Designer. The second, device control, relies on Brainflow, developed in Python, enabling cursor control through motor imagery.

Methodology and Experimentation

The pipeline begins with gathering patient information. The recording phase checks for condition-related artefacts, before the EEG is acquired and analysed to verify the ERPs linked to the motor imagery task, divided into four directional categories. Training then familiarises the user with motor imagery, preparing for the control phase, where each action is translated into cursor movement with real-time ERP transduction. and object selection through a brief closing of the eyes.

Future Developments

The project's potential extends beyond the current prototype. Future hardware points toward biopolymer electrodes covering the M1 area, connected via thinner cables to the board and communicating wirelessly with the user's device. On the software side, a single cross-platform application, available on several devices, will implement a virtual cursor controlled via neural signals, serving both clinicians, for signal analysis and remote monitoring, and patients, assisted by an integrated AI agent, with high cross-device interoperability. The result is a connection opening a new pathway: a Synapse adaptable to individual needs, and a Nexus guiding the patient toward a renewed life. Simply, SyNexus.

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